AgentSOC 2.0: An Agentic SOC Tested Against Injected Log Text and Poisoned Memory

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***Abstract*—Agentic designs for security operations centres (SOCs) put a large language model (LLM) and a learning knowledge store into the triage loop. Both consume data an attacker can shape: the model reads log text, and the store adapts to observed activity. AgentSOC, the framework we build on, was evaluated conceptually and on one proof-of-concept incident, with no adversary aimed at either. We implement its loop end to end and add a store that learns only through a learning gate, an analyst review loop with unlearning, fail-safe processing, and asymmetric trust: the LLM's output may add review but cannot dismiss a case, start an automatic action or teach the store. On 500 GUIDE v2 alert pairs, an undefended LLM agent was hijacked on 130/439 pairs, while the pipeline's response was suppressed on 4/428; these are different measures. On 14 test days of DEDALE testbed logs, learning the benign weeks cut host reviews from 6095.4 to 65.2 per day while the same 400 labelled attack events still raised alerts. Under asymmetric trust, 0/72 attack cases were dismissed in E1 on DEDALE, whichever text the attacker injected. Prompt-level defences were not tested as baselines.**

***Index Terms*—security operations, agentic AI, prompt injection, memory poisoning, alert triage**

I. Introduction

Security operations centres (SOCs) receive far more alarms than analysts can examine, and most of those alarms turn out to be harmless [5]. Agentic designs answer this by placing an LLM and a knowledge store inside the triage loop. AgentSOC [1] is such a design: an LLM suggests how an incident might unfold, a graph check removes suggestions the environment cannot support, a scoring module ranks responses, and a store keeps enterprise context together with what earlier cycles observed.

Two parts of this arrangement read data the attacker can shape. The LLM reads free text from the logs: a command line, a script block, a raw event record. Whoever runs a process on a monitored host chooses that text, and text that reaches an LLM can redirect it [6]–[8]. The store adapts to what it observes, and a detector that learns from observed traffic can be trained to accept an attack [12]; agent memories and retrieval corpora share the weakness [13], [14]. The AgentSOC paper assumed an attacker who steals credentials, moves laterally and escalates privileges, and it demonstrated the loop on a single proof-of-concept incident [1]. That scope suits the introduction of an architecture. It leaves open what happens when the attacker targets the model's input or the store's memory, and how the loop behaves over weeks of telemetry with analysts in it.

We address these questions with a working system, AgentSOC 2.0, and make three contributions:

1) An end-to-end implementation of the perception–reasoning–action loop, extended with a knowledge store that learns benign behaviour only through a learning gate, an analyst inbox whose verdicts feed the store, and unlearning once a host is confirmed compromised. We evaluate it on clean and contaminated alert pairs from GUIDE v2 and on 14 test days of DEDALE testbed telemetry.
2) Measurements of both attack surfaces: text an attacker writes into logs (on GUIDE, and through the whole loop on DEDALE), and poisoning of the learned memory, either by learning without a gate or through a misled analyst.
3) Asymmetric trust, a rule under which the LLM's output can add review but cannot dismiss a case, start an automatic action or make anything learnable. It holds by construction for what the LLM's output can cause, and each case can be checked, because the pipeline derives two decisions for every case, one that ignores the model and one that uses it, and keeps the first unless the second asks for review. We combine it with fail-safe processing, so that input the pipeline cannot handle leads to review rather than to a stalled loop.

Section II summarises the base paper and related work, Section III describes the system, Section IV the evaluation setup and Section V the results; Section VI discusses them together with the limitations.

II. Base Paper and Related Work

A. AgentSOC

AgentSOC [1] arranges SOC automation as a perception layer, an agentic reasoning layer, an action and playbook layer and an internal knowledge store, with real-time monitoring returning execution outcomes to the store. The reasoning layer has three engines. The Narrative Counterfactual Engine (NCE) prompts an LLM for several candidate attack progressions per incident, each tied to ATT&CK techniques [4] and given a confidence. The Structural Simulation Engine (SSE) walks identity, privilege and network graphs and drops candidates whose preconditions fail. RSEM, the Risk Scoring and Evaluation Module, orders candidate responses by a weighted difference of containment value and business impact. Playbooks with a high projected impact go to analysts, execution begins in dry-run mode with audit logs and rollback, and the store holds

static enterprise context next to knowledge derived from earlier cycles.

The base paper evaluates the design on a conceptual enterprise scenario and a proof of concept that used a 5,000-event sample of the LANL authentication data and a synthetic 50-node topology, with GPT-4 proposing the hypotheses; its walk-through incident took about 506 ms end to end, 480 ms of it in the LLM [1]. The authors list the fixed sample, the small topology, dry-run actions without feedback from live systems and simplified business-impact scores as limitations, and name continuous telemetry and human-in-the-loop checkpoints as future work. Three questions are open: whether data an attacker writes can steer the reasoning layer, whether the store's learning can be turned against the defender, and how the loop behaves over time with analysts. Table I sets the base design and this work side by side.

B. Related work

Prompt injection was first shown with instructions written into the prompt itself [6], then indirectly, through content that an LLM-integrated application retrieves [7]; Liu et al. formalise and benchmark attacks and defences [8]. Logs are content of that kind: the attacker produces them and the SOC's model reads them. Defences at the prompt or model level mark untrusted input so the model can tell it apart (spotlighting [9]) or train the model to follow only a separate instruction channel (StruQ [10]); they make the model follow injected text less often. CaMeL [11] instead fixes, outside the model, what untrusted data may influence. Asymmetric trust applies that second approach to a SOC: it does not try to make the model resist, it limits what the model's output can change. Poisoning of learning detectors has a long history [12], and LLM agents inherit it through memories and retrieval stores [13], [14]; we use a targeted form of unlearning [15] as one remedy. Our data come from GUIDE [2], Microsoft's incident-triage corpus, and DEDALE [3], a labelled APT dataset recorded on a testbed.

III. System

A. Pipeline

Fig. 1 shows AgentSOC 2.0. Events from endpoint and network sensors are normalised and split into two kinds of field. Structured fields (host, account, the parent process as recorded, addresses, ports, event identifiers) come from sensors and the asset inventory; we assume the attacker cannot alter them. Free text (command lines, script text, raw log records, process and file names as strings) is whatever the attacker's activity produced, so the attacker controls it.

On host telemetry, detectors for new process lineage, named pipes, PowerShell activity, writes of executable files, logons and access to the credential process compare each event's structured pattern with the knowledge store and raise an alert when the pattern has not been learned; alerts sharing a pattern form a case. For each case NCE asks the LLM for ATT&CK-labelled hypotheses, reading the first 512 characters of every free-text field. SSE checks each hypothesis against the environment model, and entity binding flags a hypothesis

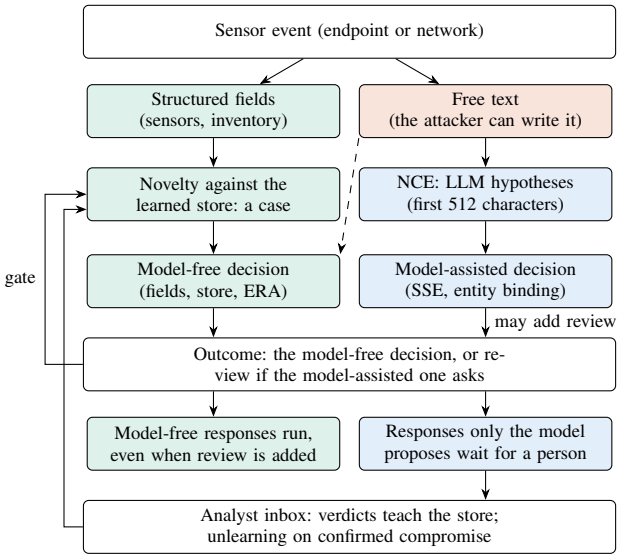


Fig. 1. AgentSOC 2.0 under asymmetric trust. Green: data and decisions the attacker cannot write; orange: free text; blue: what depends on the LLM. The dashed arrow is ERA, which reads free text and can only escalate. The gate passes to the store only cases dismissed without escalation; analyst verdicts reach it directly.

whose host or account the structured fields do not support. RSEM ranks responses, and guardrails decide which of them may run without an analyst. On GUIDE a further component, ERA (evidence risk assessment), scans free text for injection-like content; it can only escalate. Cases that need a person go to the analyst inbox, and an outcome monitor follows executed responses.

B. Asymmetric trust

In our earlier runs NCE's hypotheses fed the decision directly; we call this the *acting rule*. A hypothesis judged infeasible could let a case be dismissed, and one that survived could lead to an automatic response within the guardrails. Asymmetric trust replaces it with two decisions per case. The *model-free decision* uses everything except the LLM's output: the structured fields, the store, ERA and the check of destination telemetry. The *model-assisted decision* adds NCE's hypotheses. The outcome is the model-free decision, unless the model-assisted decision asks for review, in which case the outcome is review. Responses of the model-free decision run automatically even when review is added; a response that only the model-assisted decision proposes waits for an analyst. Only dismissed cases can be learned, and only the model-free decision can dismiss.

So, by construction for what the LLM's output can cause, the model cannot dismiss a case, trigger an automatic response or teach the store; at most it adds a review. The rule covers nothing an analyst decides after reading attacker text, and nothing about tampered sensor fields. Since it compares two decisions, it is easy to verify: our tests replay recorded model answers and randomly generated adversarial hypotheses and

TABLE I
AGENTSOC [1] AND THIS WORK. BASE-PAPER ENTRIES COME FROM ITS SECTIONS III AND IV; ENTRIES FOR THIS WORK NAME THE SECTION OF OUR RESULTS FILE (C1–C5).

Aspect	AgentSOC (base paper)	This work
Attacker model	Steals credentials, moves laterally, escalates privileges; detecting zero-days and analysing malware are out of scope	Adds an attacker who writes text into logs the LLM reads, and one who poisons the learned memory (C1, C3)
Memory and learning	Store keeps enterprise context and knowledge updated from perception and action cycles	Store learns benign patterns only through a learning gate; unlearning after a confirmed compromise (C2, C3)
Analyst loop	High-impact playbooks routed to analysts; human-in-the-loop checkpoints listed as future work	Analyst inbox whose verdicts teach the store; response delays from 1 to 72 hours measured (C3, C5)
LLM output	SSE removes structurally infeasible hypotheses	Asymmetric trust: by construction, for what the LLM's output can cause, the output can only add review (C1, C4)
Failure handling	Dry-run start, audit logs and rollback paths	This work adds: malformed events refused and counted; failing events quarantined, with recovery from the last checkpoint; pipeline errors routed to review
Data	5,000-event sample of LANL authentication logs; synthetic 50-node topology	GUIDE v2: 500 pairs for development, 500 held out; DEDALE testbed host and network logs (C1, C2)
LLM	GPT-4 in the proof of concept	gemini-3.1-flash-lite; gpt-oss-20b on the GUIDE development pairs (C1)
Evaluation	Conceptual scenario and one proof-of-concept incident (about 506 ms end to end)	Paired contamination tests; 14 test days under learning, analyst and LLM conditions; Wilson intervals and exact McNemar tests (C1–C5)

check that no outcome is less strict than the model-free decision.

C. Learning gate and unlearning

The store learns a case's pattern only when the case was dismissed without any escalation. Under asymmetric trust that means dismissed by the model-free decision, so neither attacker text nor the model can open the gate. Analyst verdicts are the deliberate exception: a benign verdict teaches the store directly, which is how feedback lowers the workload. When an analyst confirms a host as compromised, the store retracts what it recently learned on that host and learns nothing further there.

D. Fail-safe processing

A loop that an input can stall serves the attacker as well as a loop that an input can fool. Malformed events are refused and counted instead of being passed on. An event on which the pipeline fails is moved to a quarantine file, and the live runner restores its last checkpoint and carries on; before this change such an event ended the loop. An error inside triage sends the case to review instead of dropping it, and analyst commands are applied atomically, so a failing command leaves no partial change. Checkpoints are signed and loaded through an allow-list of classes. A check that scans evidence text for command-and-control indicators reads only the part of each field the LLM was shown, which bounds its running time on long crafted text.

IV. EVALUATION SETUP

A. GUIDE v2: contaminated alert pairs

From GUIDE [2] we built pairs of alerts that differ at a single injection site: a clean alert and a contaminated twin carrying an attacker payload, with 500 pairs for development and 500 held out (C1). Entities, evidence text and payloads are synthetic. GUIDE alerts run through the single-alert pipeline without a learned store, so NCE's hypotheses are what raise a response there. A pair is *suppressed* when the clean alert received a response (review or an automatic response) and its twin did not. We compare suppressed pairs with flips in the opposite direction using an exact McNemar test, keeping apart the flips caused by ERA's escalation, which are the defence reacting; proportions carry 95% Wilson intervals. The undefended comparison agent is the same model, prompted with the alert's fields to return a verdict; it is *hijacked* when it does not dismiss the clean alert but dismisses the twin. Suppression concerns the pipeline's decision and hijacking the agent's own verdict, so the two are different measures. We also count *automatic containments removed*: pairs whose clean alert runs an automatic block the trusted telemetry supports while the twin does not.

B. DEDALE: weeks of testbed telemetry

DEDALE [3] records an emulated small company over several weeks, including one multi-step attack; exact labels exist for the events of the 2 attacked Windows clients. The store learns from the two benign weeks, and the 14 test days that follow are replayed in order, with host logs and network (Suricata) alerts (C2). A simulated analyst answers from the labels after a set delay, optionally with reproducible mistakes. In experiment E1 the attacker appends text to the command lines of their own processes: a routine-maintenance claim, a fake SOC approval, or a request to isolate the domain controller instead; 84 events carry it, and in 38 cases the first alert does (C1).

C. Recorded answers, two rules, frozen results

Every LLM answer was recorded once (gemini-3.1-flash-lite on both datasets, gpt-oss-20b on the GUIDE development pairs) and is replayed, so no experiment needs a model call to

TABLE II

ATTACKER-WRITTEN TEXT UNDER THE ACTING RULE AND UNDER ASYMMETRIC TRUST (C1; ASYMMETRIC-TRUST ROWS OF C1). GUIDE: PAIRS; E1 ON DEDALE: ATTACK CASES. NO STEERED ACTION WAS EXECUTED IN ANY RUN.

Setting	Measure	Acting rule	Asym. trust
GUIDE dev, Gemini	suppressed	4/428	4/428
	containments removed	20/178	0/178
GUIDE held-out, Gemini	suppressed	3/437	3/437
	containments removed	18/168	0/168
GUIDE dev, gpt-oss-20b	suppressed	10/437	10/437
	containments removed	21/176	0/178
E1 on DEDALE, no analyst	dismissed, maintenance	1/72	0/72
	dismissed, approval	3/72	0/72
	dismissed, steering	1/72	0/72
	reviews per day, clean	54.8	65.2
E1 on DEDALE, analyst (4 h)	dismissed, each text	0/72	0/72
	reviews per day, clean	33.2	34.1

reproduce. Each condition is a deterministic replay, and one script rebuilds our results file from the frozen outputs; every number below is taken from that file, and we name its section (C1–C5). The acting rule and asymmetric trust are scored on the same recorded answers. The asymmetric-trust numbers are the rows marked as such in sections C1 and C4 of the results file.

V. RESULTS

A. C1: text written into logs

On GUIDE, contamination suppressed the pipeline's response on 4 of 428 development pairs whose clean alert was answered (0.9%, 95% CI 0.4%–2.4%), on 3 of 437 held-out pairs (0.7%, 0.2%–2.0%), and on 10 of 437 development pairs when gpt-oss-20b produced the hypotheses (2.3%, 1.2%–4.2%). Against the flips in the other direction not caused by ERA, McNemar gives $p = 1$, 0.25 and 0.3: no detectable effect. ERA's escalation produced the opposite flip in 12, 10 and 12 pairs. No steered hypothesis led to an executed action; 7, 2 and 3 were held for an analyst. The undefended agent was hijacked on 130 of 439 pairs (29.6%, 25.5%–34.0%, $p = 6.5 \times 10^{-17}$). In E1 on DEDALE, under the acting rule and without an analyst, the maintenance text caused 1 dismissal among the 72 attack cases, the approval text 3 and the steering text 1; with an analyst, there were none. In every E1 run the store learned no attack pattern and the same 400 labelled attack events raised alerts. The steering text did reach the model: NCE named the steered host in 30 of 38 injected cases (32 of 38 with the analyst), entity binding flagged every steered hypothesis (49 of 49 and 59 of 59), and no containment targeted that host.

B. C2: learning behind a gate

With nothing learned, the host detectors produced 6095.4 reviews per day over 85336 test cases, raising 80 malicious cases and 473 labelled attack events (C2). After learning the benign weeks, reviews fell to 65.2 per day over 913 cases; 72 malicious cases were raised, every one reviewed, and 400 labelled attack events alerted. Continuing to learn behind the gate gave the same numbers: with neither NCE nor an analyst, no case is dismissed, so the gate admits nothing new, and the reduction is due to the training weeks rather than to the gate. Learning without the gate is where the memory turns against the defender. Reviews dropped to 27.4 per day, but the store absorbed 38 attack cases' patterns as routine, only 38 malicious cases were raised, and 99 attack events alerted. Network alerts followed the same pattern: 552.3 reviews per day with nothing learned, 26.1 after learning, and without the gate 20.6, with 1 attack pattern learned.

TABLE III

DEDALE HOST LOGS, 14 TEST DAYS. TOP: LEARNING AND POISONING (C2, C3); BOTTOM: THE LLM (C4; THE ASYMMETRIC-TRUST ROWS OF C4). PATTERNS: ATTACK CASES' PATTERNS LEARNED AS ROUTINE.

Condition	Reviews/day	Malicious raised	Events alerted	Patterns
Nothing learned	6095.4	80	473	0
Benign weeks learned, gate	65.2	72	400	0
Learning without the gate	27.4	38	99	38
Analyst misses 20%	30.1	61	256	20
with unlearning	34.1	72	400	0
	Reviews/day	Malicious dism.	Benign dism.	
NCE, closed world	54.6	2/72	89	
NCE, open world	54.8	0/72	89	
NCE and analyst	33.2	0/72	30	
NCE, asymmetric trust	65.2	0/72	0	
NCE and analyst, asym. trust	34.1	0/72	0	

C. C3: poisoning through the analyst

The gate does not apply to analysts, so we simulated one who misses attacks (C3). Missing 5% let the store learn 8 attack cases' patterns and left 397 events alerting; missing 20% taught it 20 patterns and reduced the malicious cases raised to 61 and the alerting events to 256. With unlearning on confirmed compromise, both mistake rates ended with 0 patterns learned, 72 malicious cases raised and 400 events alerting. The price is review load: 34.1 instead of 30.9 reviews per day with an analyst who is always right, and 44.6 instead of 31.4 with one who raises 5% false alarms, since every false confirmation retracts benign learning.

D. C4: the LLM on testbed telemetry

Adding NCE under the acting rule with a closed-world feasibility check lowered reviews from 65.2 to 54.6 per day, but dismissed 2 of 72 malicious cases together with 89 benign ones (C4). Both misses involved a server without telemetry in the store, whose absence was read as impossibility. Treating unmonitored hosts as unknown (open world) dismissed none of the 72 malicious cases and 89 benign ones, at 54.8 reviews per day; we adopted this rule after seeing the misses, on the same data. With an analyst as well, NCE brought reviews to 33.2 per day against 34.1 for the same analyst without NCE (C3), dismissing 30 benign cases. The workload NCE saves comes entirely from dismissing cases, the same route the attacker used in E1. Each free-text field reaches the model cut to its first 512 characters; the command line exceeds that length in 1 of 797 first alerts, the raw log line in 718 of 856.

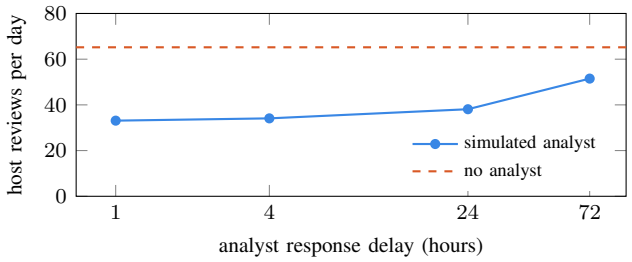


Fig. 2. Host reviews per day against the analyst's response delay (C5). At every delay the 72 malicious cases were raised and the same 400 labelled attack events alerted.

E. C5: running the analyst loop

Analyst feedback lowers the load further, and its delay matters (Fig. 2, C5): 33.1, 34.1, 38.1 and 51.5 reviews per day with answers after 1, 4, 24 and 72 hours, against 65.2 without an analyst, with 72 malicious cases raised and the same 400 events alerting at every delay. On the network side the attacker's server was contained after 60, 240, 1440 and 4320 minutes, the analyst's delay in each case, and never without an analyst. With a 4-hour analyst every stage of the attack reached review: initial access on client 2 first after 1.2 minutes (20 cases), lateral movement to client 1 after 28.5 (11), collection from the internal server after 30.0 (11) and exfiltration after 1265.0 minutes (4). With 5% false alarms on the network side, approved blocks would have hit benign traffic 8 times.

F. C6: asymmetric trust

We re-scored the same recorded answers under asymmetric trust; these numbers are the asymmetric-trust rows of C1 and C4 (Tables II and III). On GUIDE, suppression did not change (4/428, 3/437, 10/437): those clean alerts were answered only because the model read their clean text, and limiting the model's output cannot restore what the injected text hid. The rule removed a different effect. Under the acting rule any review held back an alert's automatic responses, so contamination that triggered a review cancelled a block the trusted telemetry supported: in 20 of 178 (11.2%), 18 of 168 (10.7%) and 21 of 176 (11.9%) pairs. ERA's own, correct escalation was the trigger in 17, 18 and 17 of these pairs. Under asymmetric trust the block runs and the review is added: 0 of 178, 168 and 178. The denominators can grow because a clean alert's block now also runs when its own hypotheses ask for review. Outcome totals on the development pairs stayed at 532 reviews and 336 automatic responses. In E1 on DEDALE, 0/72 attack cases were dismissed for every injected text, with and without an analyst; no pattern was learned and the same 400 events alerted. Reviews per day were 65.2 without and 34.1 with an analyst whatever the injected text, so the text no longer changes the workload. The cost is the dismissals NCE used to make: without an analyst the system reviews 65.2 cases a day instead of 54.8; with one, 34.1 instead of 33.2.

VI. Discussion and Limitations

Robustness came from structure, not from the model. On its own the model was easy to mislead: as an undefended agent it was hijacked on 130 of 439 GUIDE pairs (29.6%), and in E1 it pointed at the steered host in 30 of 38 injected cases (C1). What kept injected text from acting was the decision structure around the model, the principle CaMeL [11] applies to agents in general; asymmetric trust turns it into a rule that can be checked case by case.

Limits on output do not help where the model is the detector. GUIDE's 4 of 428 suppressed pairs remain under asymmetric trust because the model's reading of the clean text was the only thing that raised them. On DEDALE the store's novelty check raises the cases and the model only adds to them, which is why 0/72 attack cases were dismissed in E1 on DEDALE. Detection should rest on signals the attacker does not write.

Review should be added to automatic containment, not substituted for it. The containment result (C6) shows that a correct escalation becomes the attacker's lever when it also pauses a response that trusted evidence already supports.

The analyst is the open channel. Analysts have to read the evidence, attacker text included, and their benign verdicts teach the store. C3 shows how far a misled analyst can poison it; unlearning helps only after some analyst confirms the host, which happened for both attacked hosts in our runs.

Limitations.

- One testbed dataset with one attack scenario, and labels for 2 attacked Windows clients (the results file's Limits section). In GUIDE v2, the entities, evidence text and payloads were generated synthetically.
- A simulated analyst: an oracle that follows the labels, with independent random mistakes. Real analysts' errors are likely to cluster on the cases an attacker disguises best, which would likely make the poisoning results in C3 worse.
- DEDALE was run with a single model, gemini-3.1-flash-lite; gpt-oss-20b was used on the GUIDE development pairs only. Each condition ran once; replay reproduces the same numbers on every rebuild, but we did not measure how much the model's answers vary between runs.
- The analyst channel: asymmetric trust covers what the LLM's output can cause, not what an analyst decides after reading attacker text. We also assume the structured sensor fields are not tampered with; an attacker who can alter them is out of scope.
- Each free-text field reaches the model cut to 512 characters. On DEDALE the command line exceeds that in 1 of 797 cases, while the raw log line exceeds it in 718 of 856, so attacker text placed late in a raw line never reaches the model (C4). With asymmetric trust in place, the loss is at most an escalation that does not happen.
- On GUIDE, where the model is the only detector, contamination still suppressed 4 of 428 responses, with or without asymmetric trust (C1).

- Attack activity whose structured pattern also occurs in benign activity stays silent: of the 473 labelled attack events that raise alerts before any learning, 73 no longer do once the benign weeks are learned (C2). Many labelled events are of kinds the detectors do not watch.
- No prompt-level defence [9], [10] was evaluated as a baseline: the pipeline is compared only with an undefended triage agent. A comparison with a hardened prompt, in which the untrusted fields are delimited and declared to be data, was prepared on DEDALE but could not be completed, because the model's API returned only overload errors (HTTP 503) during the attempts.
- The host detectors and the open-world rule were chosen on DEDALE itself; E1 injects text only into command lines; all containment is simulated.

VII. Conclusion and Future Work

AgentSOC proposed combining an LLM's hypotheses with structural checks and risk-ranked responses. Building the loop and attacking it showed that the dangerous paths run through what the model's output may trigger and what the store may learn. Learning the benign weeks cut host reviews from 6095.4 to 65.2 per day with the same 400 attack events alerting (C2); without the gate the store learned 38 attack cases' patterns, and unlearning undid the poisoning a mistaken analyst caused (C3). Asymmetric trust closed the attacker's route through the model for what the LLM's output can cause: in E1 on DEDALE, 0/72 attack cases were dismissed whatever text the attacker wrote (C1). Next steps are the hardened-prompt baseline we prepared but could not complete, evaluation with real analysts, a second labelled dataset, repeated model runs, and defences for the analyst channel.

Data and Code Availability

[TODO: repository link]. The recorded LLM answers and the script that rebuilds every number are part of it.

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